

# HEURISTECH INNOVATION LABS PRIVATE LIMITED

## Coding Assessment

### Assessment Instructions:

We are excited to see your skills in action! Below are the project options for your assessment. You may choose any one of the three projects to complete within 5-7 days.

- **Preferred Language:** Flutter/Dart (Other languages are acceptable if necessary).
- **Clarifications:** If you need any clarification or wish to set up a meeting for further understanding midway through the project, please reply to this email.
- **Submission:** Submit the project with your best effort, even if it's not fully complete. Aim to get a conclusive working outcome.
- **Evaluation:** You will be evaluated based on functionality, code quality, UI/UX design, completeness of the project, and your ability to explain your work.

### Project Choices:

#### Project 1: Onboarding + Profile (Resume Builder)

##### Objective:

Build a user onboarding flow that collects structured data and displays it as a profile.

##### Requirements:

- **Authentication:**
  - Email/Password or Google Sign-In
- **Multi-step onboarding form:**
  - Personal details (name, email)
  - Skills (list input)
  - Experience (role, company, duration)

- Education
  - Interests / goals
- Store all data in Firebase Firestore with proper structure
- **Profile screen:**
  - Display all collected data in a clean format
- **Edit functionality:**
  - User should be able to update any section

### **Expected Focus:**

- Form handling and validation
- Clean UI/UX
- Proper Firestore data modeling
- State management

## **Project 2: Community Feature**

### **Objective:**

Build a basic community system where users can interact.

### **Requirements:**

- **Community creation:**
  - Name, description
  - Visibility: Public or Private
- **User actions:**
  - Join/leave community
- **Posts:**
  - Create text-based posts
  - View list of posts (feed)
- **Basic structure:**

- Show communities list
- Open a community to view posts

**Bonus (Optional):**

- Roles (Admin/User)
- Real-time updates using Firestore streams
- Simple search or filtering

**Expected Focus:**

- Data modeling (users, communities, posts)
- Real-time updates
- Scalable structure
- Clean UI

**Project 3: Voice + Multilingual Interaction (Voice-Based Onboarding)**

**Objective**

This option is similar to Option 1 (Onboarding + Profile), but instead of only typing, users should be able to complete the onboarding process using voice.

The goal is to test how well candidates can integrate voice features into a real product flow.

**Requirements**

**1. Onboarding Flow (Same as Option 1)**

- Create a simple question-based flow:
  - Example:
    - What is your name?
    - What are your skills?
    - Tell us about your experience
- Answers should be stored and shown in a profile screen

## **2. Voice Input (Speech-to-Text)**

- User should be able to:
  - Tap a microphone button
  - Speak their answer
  - App converts speech → text and fills the input field
- User should still have the option to edit typed text

## **3. Voice Output (Text-to-Speech)**

- Each question should be:
  - Displayed as text
  - Also read aloud automatically
- When moving to the next question, it should speak again

## **4. Multilingual Support**

- Support at least:
  - English
  - +1 regional language (Hindi or Telugu)
- User should be able to:
  - Select language at the start
- Based on selected language:
  - Questions should be shown and spoken in that language
  - Voice input should work accordingly

## **5. UI Requirements**

- Show clear states:
  - Recording started / stopped
  - Listening indicator
- Provide:
  - Toggle between typing and speaking

- Display recognized text in real-time or after recording

### **Expected Flow (Example)**

1. User selects language (e.g., Telugu)
2. App asks (via text + voice): “What is your name?”
3. User taps mic and speaks
4. App converts speech to text and displays it
5. User confirms or edits
6. Move to next question

### **Expected Focus**

- Proper integration of:
  - Speech-to-text
  - Text-to-speech
- Handling:
  - Permissions (mic access)
  - Errors (no speech, unclear audio, etc.)
- Smooth UX:
  - No lag between speaking and result
- Language handling:
  - Switching and consistency